

Additional examples of exam questions

Question 1

1) The simplest crop growth model originates from measurements showing that for many crops the potential growth rate amounts to $200 \text{ kg ha}^{-1} \text{ d}^{-1}$.

As you know, the Lintul model is based on the assumption that light intensity exponentially decreases with the accumulated LAI in a crop canopy, and that light can directly be related to the increase of biomass through the Light Use Efficiency (LUE).

Show that the Lintul model approximates the value of $200 \text{ kg ha}^{-1} \text{ d}^{-1}$ for a closed canopy, assuming that the light extinction coefficient, k , amounts to 0.6, the $\text{LUE}=3$, and the daily total radiation, DTR , equals $15 \text{ MJ m}^{-2} \text{ d}^{-1}$.

Question 2.

A research paper reports about data of two cultivars of spring barley. One result concerns development rates plotted versus temperature, Figure 1.

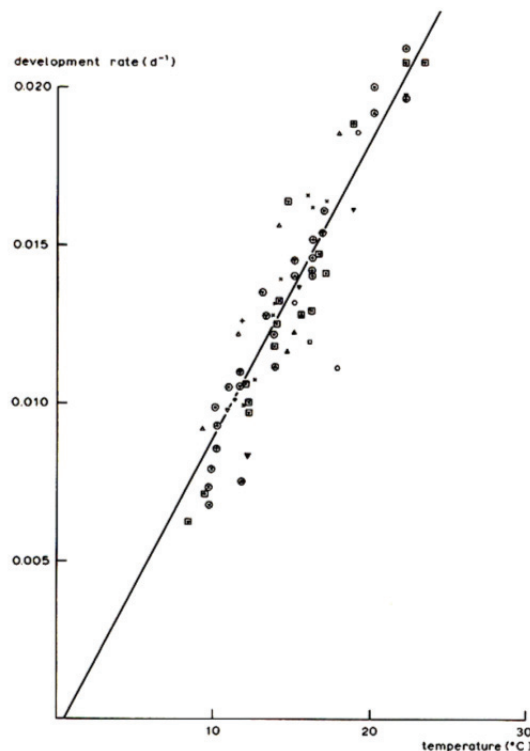


Figure 1. Development rate of barley as a function of temperature from emergence till flowering.

- 2a) Explain how the development rate, the values on the y-axis in this figure, is experimentally derived; also give the equation.
- 2b) (3.5 pts) Calculate (in an accurate way, so using a ruler for example) from Figure 1 how long it takes in terms of physiological time, so expressed in °Cd, to reach flowering.